Comprehensive Benchmark of Homography Methods for Thermal Photovoltaic Inspection: A Colormap Analysis

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***Abstract*—**
Thermal homography estimation supports key inspection tasks in photovoltaic systems, including temporal fault tracking, aerial mosaicking, and multi modal image fusion. Thermal images contain uniform temperature patterns with low contrast and limited texture. These properties differ from RGB imagery that dominates existing homography estimation benchmarks. One strategy to compensate for the limited structure in thermal data is the use of colormaps that amplify subtle temperature gradients and improve fault visibility in photovoltaic panels. Existing studies provide limited evaluation of homography methods on thermal imagery and do not examine the effect of colormap selection on performance. This paper introduces the first systematic benchmark of 22 homography estimation methods (15 feature-based and 7 deep learning-based methods) with 6 metrics on thermal photovoltaic images. The benchmark also evaluates 29 colormaps and provides a complete comparison under controlled settings. The results show a performance gap between detector-free vs detector-based models, and conventional vs deep learning methods. Detector free models achieve strong performance with success rate up to 97.09% by removing the dependence on corner structure. Deep learning methods surpass traditional approaches by 56.2% in success rate. Colormap selection produces large performance variations for traditional methods, while deep learning models maintain more consistent behavior and show reduced sensitivity to visual changes. Nevertheless, traditional pipelines remain valuable in GPU constrained and resource-limited environments. Altogether, the benchmark framework advances thermal image analysis for photovoltaic applications and provides a foundation for reliable deployment in industrial inspection.

***Index Terms*—Thermal imaging, photovoltaic inspection, homography estimation, deep learning, feature-based methods, colormap analysis, benchmark dataset, renewable energy, THED-PV.**

I. Introduction

Photovoltaic (PV) systems form a critical part of global energy infrastructure, with installed capacity reaching 1 419 GW by 2023 and expected to exceed 2 300 GW by 2027 [1], [2]. Thermal imaging is the primary technique for identifying faults in PV panels and reveals hotspots, microcracks, and electrical failures that cannot be observed through visual inspection [3]–[5]. The shift from manual surveys to automated drone based inspection of large scale PV farms increases the need for accurate alignment of thermal images so that faults can be detected with consistency across time and location [6], [7].

Homography estimation is crucial for aligning thermal images in automated PV inspection workflows. It enables consistent temporal fault tracking by registering sequential images, making it possible to detect changes in hotspots and degradation over time [8], [9]. Image alignment is particularly important because analyzing individual thermal images separately can miss critical patterns of temporal evolution or overlook subtle changes in panel performance that only become apparent when images are compared over time [10], [11]. Additionally, homography supports the creation of large-scale thermal mosaics, where overlapping aerial images of thousands of panels are stitched together, providing a comprehensive view of the PV system [12], [13]. It also enables multi-modal fusion, correlating thermal anomalies with visual data to assist in precise fault localization [14], [15]. Despite these capabilities being central to effective PV monitoring, most practitioners still rely on homography methods designed for RGB images, often without validating their applicability to the unique characteristics of thermal imagery.

Thermal imagery of PV panels exhibits visual properties that diverge sharply from those found in natural image domains. Panels that operate under normal conditions present nearly uniform temperature fields with variations that remain between 2° and 5°C across individual cells [3], [5]. This behavior produces broad regions with weak gradients and minimal local texture. The infrared response of silicon introduces reflective artifacts, and ambient conditions create additional thermal fluctuations [9], [16], [17]. These factors reduce the availability of stable geometric structure and limit the consistency of local features that conventional detectors typically exploit [18], [19].

Classical detectors return far fewer keypoints and their outputs display lower repeatability and weaker descriptor separability compared to natural image counterparts [20], [21]. Gradient statistics confirm this discrepancy as average gradient magnitudes decrease by approximately 65% in thermal PV images relative to visually comparable color

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scenes [22]. These findings reveal a structural mismatch between the assumptions behind established feature based pipelines and the properties of thermal data. They therefore raise concerns about whether performance reported by homography benchmarks on natural images [23], [24] can be used to infer behavior in environments where texture, gradients, and reflectance conditions follow a fundamentally different distribution. This limitation motivates a closer examination of the benchmarks themselves and the extent to which they represent the visual characteristics encountered in thermal PV inspection.

Existing homography benchmarks provide limited guidance for thermal PV inspection. Studies that examine thermal registration address medical imaging [25], [26] or surveillance scenarios [27], [28] and do not reflect the visual characteristics of PV panels. Research on thermal analysis in PV systems concentrates on fault classification and often treats image alignment as a solved preprocessing step [29]–[31]. These gaps leave practitioners without clear guidance on selecting suitable homography methods for the analysis of PV panels.

Thermal cameras record radiometric measurements as single channel intensity, yet these intensities are routinely transformed through color mapping functions to support improved visual interpretation [32], [33]. A colormap can be expressed as a function

$$C : [0,1] \to [0,1]^3, \qquad C(I) = \big(R(I), G(I), B(I)\big),$$

where I denotes the normalized thermal intensity and the output represents a triplet of color values. This transformation assigns perceptually meaningful colors to small temperature variations and produces local contrast that is often absent in the raw grayscale representation. Perceptually uniform schemes facilitate accurate interpretation of thermal differences, while sequential colormaps preserve ordered value relationships and maintain clarity on PV surfaces that contain limited texture [34], [35]. These properties amplify weak gradients, expose structural cues, and reveal spatial patterns that would otherwise remain indistinct. Although colormaps are widely used in PV inspection pipelines, their influence on homography estimation has not been investigated.

This study advances homography estimation for thermal PV inspection by analyzing how color map choices and environmental factors influence homography estimation. The contributions are summarized below:

- A comprehensive benchmark for homography estimation on thermal PV imagery is introduced. It evaluates 22 categorized methods, including detector based and detector free pipelines, across 99,680 image pairs.
- A systematic evaluation of twenty nine colormaps is conducted. The colormaps are organized into a taxonomy that enables a structured assessment of how representation choices affect feature visibility and homography.
- Homography performance is analyzed across different times of day and diverse environmental conditions to characterize the sensitivity of existing methods to real world thermal variations.
- A broad set of accuracy and efficiency metrics is reported and the findings are consolidated into practical recommendations for reliable thermal PV inspection.

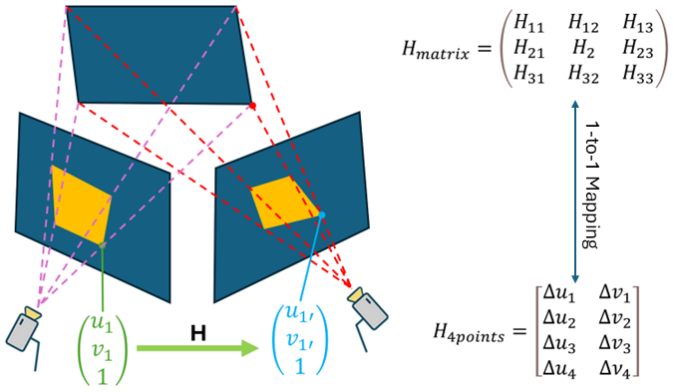


Fig. 1. Homography estimation framework for planar surface alignment.

This work extends our conference study [20] from 3 methods on limited data to 22 approaches across all major paradigms, expands colormap analysis from 22 to 29 variants, employs 7 metrics for comprehensive assessment, and provides deployment guidelines absent from conference work. Section II establishes homography mathematics and thermal imaging principles. Section III reviews estimation methods and prior benchmarks. Section IV details experimental methodology. Section V presents results and analysis. Section VI provides industry guidelines and discusses implications. Section VII summarizes contributions.

II. Homography Estimation: Mathematical Framework

A homography defines the projective transformation that relates two views of a planar surface [36]. The mapping is represented by a 3×3 matrix $\mathbf{H}$ that transforms points between image planes through homogeneous coordinates. For two corresponding points $\mathbf{p} = [u, v, 1]^\top$ and $\mathbf{p}' = [u', v', 1]^\top$, the relationship is expressed as

$$\mathbf{p}' \simeq \mathbf{H}\,\mathbf{p},$$

where $\simeq$ denotes equality up to scale. The matrix $\mathbf{H}$ contains eight independent parameters and is typically normalized by enforcing $h_{33} = 1$ or $\|\mathbf{H}\| = 1$.

Estimating $\mathbf{H}$ begins with establishing point correspondences between the two images. A minimum of four correspondences $\{\mathbf{p}_i \leftrightarrow \mathbf{p}'_i\}$ is required because each pair contributes two independent linear constraints. The direct linear transformation (DLT) algorithm converts the relation $\mathbf{p}' \simeq \mathbf{H}\mathbf{p}$ into a set of linear equations in these unknowns and assembles them into a homogeneous system $\mathbf{A}\mathbf{h} = \mathbf{0}$, where $\mathbf{h}$ stacks the nine elements of $\mathbf{H}$. The solution is obtained by selecting the right singular vector of $\mathbf{A}$ that corresponds to the smallest singular value of $\mathbf{A}^\top\mathbf{A}$ (rayleigh quotient minimization principle).

TABLE I
COMPARISON OF TRADITIONAL KEYPOINT DETECTION AND DESCRIPTION METHODS

Method	Year	Detector Type	Descriptor Size	Scale Inv.	Rotation Inv.	Speed	Key Characteristics
SIFT [18]	2004	DoG	128-D float	Yes	Yes	Slow	Robust baseline; gradient dependence limits thermal performance
ORB [19]	2011	FAST	256-bit binary	No	Yes	Fast	Real-time efficiency; corner detection fails in uniform regions
BRISK [41]	2011	AGAST	512-bit binary	Yes	Yes	Fast	Scale-invariant with binary efficiency
KAZE [42]	2012	Nonlinear	64-D float	Yes	Yes	Medium	Nonlinear diffusion preserves weak thermal boundaries
AKAZE [43]	2013	Nonlinear	486-bit binary	Yes	Yes	Medium	Accelerated KAZE with binary descriptor

Correspondences extracted from real imagery often include outliers that arise from detection noise, descriptor ambiguities, and mismatched features. Robust estimation therefore relies on consensus-driven strategies. The RANSAC algorithm [37] repeatedly samples minimal four-point subsets, computes provisional homographies through DLT, and evaluates each hypothesis by counting inliers whose reprojection error remains below a threshold. The hypothesis with the largest support is then refined with all inliers to obtain the final estimate. Extensions such as PROSAC [38], MLESAC [39], and MAGSAC [40] enhance this process through guided sampling, probabilistic scoring, and adaptive thresholds, which improve both stability and accuracy in challenging conditions.

III. Related Works

This section reviews conventional and deep learning-based homography estimation methods.

A. Traditional Keypoint-Based Methods

Classical homography estimation relies on detection, description, matching, and robust geometric fitting. These methods differ in how they construct scale space, encode gradient structure, and balance accuracy with computational efficiency. Table I summarizes reviewed methods with thermal imaging considerations.

SIFT [18] detects scale-space extrema using Difference of Gaussians and describes local patches with gradient orientation histograms. Its descriptors remain highly discriminative but depend strongly on gradient strength, which reduces stability on low contrast thermal imagery. ORB [19] combines FAST corner detection with rotation-aware binary descriptors and offers real time performance, though the absence of scale invariance and its reliance on corner intensity contrasts limit performance on thermally uniform surfaces. BRISK [41] introduces a scale space built from AGAST detectors and produces 512 bit binary descriptors that improve robustness while maintaining efficiency. KAZE [42] and AKAZE [43] construct nonlinear diffusion scale spaces that preserve weak gradients more effectively than Gaussian blurring, which makes them better suited for subtle thermal structures found near fault boundaries.

B. Deep Learning-Based Homography Estimation

Deep learning approaches replace hand-crafted stages with learned detection, description, matching, or direct regression. These methods differ in how they leverage supervision, geometric constraints, and global context. Table II compares deep learning-based homography approaches.

HomographyNet [44] introduced direct regression of the eight homography parameters using convolutional encoders trained on synthetic data. While effective on natural scenes, its reliance on RGB training limits generalization to thermal imagery. Learned local feature pipelines extend classical designs. SuperPoint [45] uses homographic adaptation to generate repeatable keypoints, and SuperGlue [46] employs attention based graph neural networks for context aware matching. D2-Net [47] jointly optimizes detection and description, while R2D2 [48] models repeatability and reliability explicitly. DISK [49] learns an end-to-end policy through reinforcement learning.

Detector-free models eliminate keypoint extraction entirely. LoFTR [50] matches dense features using transformer-based coarse to fine reasoning and captures long range context that remains valuable in low texture thermal regions. Hybrid designs extend these ideas. LightGlue [51] adapts computation to match difficulty and XFeat [52] targets high efficiency for aerial and embedded inspection. These methods illustrate the shift toward flexible, context driven correspondence estimation that may offer advantages in thermal PV applications.

IV. Experimental Methodology

This section details our experimental methodology, including the THED-PV dataset, the 29 colormap and preprocessing variants representing our central contribution, the 22 evaluated methods, and comprehensive evaluation metrics.

A. THED-PV Dataset

THED-PV is a large-scale PV thermal imagery dataset for benchmarking and developing homography estimation methods [53]. It contains 12,460 thermal images captured at 640×512 resolution using a long-wave infrared camera under controlled variations in camera height, viewing angle, capture time, and surface conditions. Data collection

TABLE II
COMPARISON OF DEEP LEARNING-BASED HOMOGRAPHY ESTIMATION METHODS

Method	Year	Category	Supervision	GPU	Key Innovation & Thermal Relevance
HomographyNet [44]	2016	Direct Regression	Supervised	Yes	First CNN homography estimator; RGB training limits thermal transfer
SuperPoint [45] + SuperGlue [46]	2018/2020	Detector-based	Self+Sup.	Yes	Self-supervised detection with GNN matching; global context
D2-Net [47]	2019	Detector-based	Supervised	Yes	Joint detection-description; dense extraction
R2D2 [48]	2019	Detector-based	Supervised	Yes	Explicit repeatability and reliability learning
DISK [49]	2020	Detector-based	RL-based	Yes	End-to-end RL optimization
LoFTR [50]	2021	Detector-free	Supervised	Yes	Dense transformer matching; designed for low-texture
LightGlue [51]	2023	Hybrid	Supervised	Yes	Adaptive matching with early termination
XFeat [52]	2024	Full Pipeline	Supervised	Yes	Extreme efficiency for real-time applications

occurred over five consecutive days at an outdoor PV test facility and spans a range of irradiance levels and environmental states representative of real PV inspection scenarios.

Synthetic homography pair generator provided with the dataset builds on the protocol introduced by DeTone et al. [44]. Corner perturbations and controlled translations generate image overlaps that approximate close-range camera motion. The dataset also incorporates a standardized preprocessing pipeline designed to address challenges specific to thermal PV imagery, including low texture, glare, shadows, and radiometric noise [53]. The preprocessing pipeline improves feature detectability and produces images with consistent geometric structure suitable for homography evaluation. Additional acquisition details are available in the accompanying dataset report [53].

B. Colormap Representation Analysis

Thermal cameras capture single-channel radiometric data, but inspection software applies false-color mappings for visualization and processing [32], [33]. A central objective of this study is to examine how colormap selection influences homography estimation. This aspect has not been investigated in prior work, even though colormaps are routinely chosen in thermal inspection and fundamentally alter image structure by introducing artificial gradients through their color progressions [34], [35].

We evaluate 27 colormaps organized into five categories: *Sequential* (`Blues`, `Reds`, `Oranges`, `Greys`) provide single-hue progressions; *Diverging* (`BrBG`, `RdBu`, `PiYG`, `coolwarm`) employ two-sided scales with neutral midpoints creating strong gradients; *Perceptually uniform* (`viridis`, `plasma`, `inferno`, `magma`, `cividis`) ensure consistent perceived intensity changes [35]; *Qualitative* (`Set1`, `Set2`, `Pastel1`, `tab10`) use discrete categorical hues; *Specialized thermal* (`cool`, `copper`, `hot`, `nipy_spectral`, `seismic`, `terrain`, `twilight`, `twilight_shifted`) represent diverse thermal imaging practices.

For compatibility with single channel architectures, the three channel outputs are converted to grayscale using the luminance projection

$$I_{\text{gray}} = 0.299R + 0.587G + 0.114B,$$

which preserves colormap induced gradient structure. We include raw thermal data as the 28th variant. Combined with the preprocessing pipeline (Section IV-A), this yields 29 total representations per base pair, enabling systematic analysis of colormap impact on homography estimation across traditional and deep learning methods.

C. Evaluation Metrics

We evaluate accuracy, robustness, and computational efficiency using five complementary metrics that follow established homography benchmarking practices [23], [54].

Mean Average Corner Error (MACE) measures geometric accuracy by comparing the reprojection of image corners under the ground truth and estimated homographies. For corners $\mathbf{c}_i$,

$$\text{MACE} = \frac{1}{4}\sum_{i=1}^{4} \|\mathbf{H}_{\text{gt}}\mathbf{c}_i - \mathbf{H}_{\text{est}}\mathbf{c}_i\|_2 .$$

Lower values indicate tighter alignment.

Root Mean Square Error (RMSE) applies a quadratic penalty to large deviations and complements MACE by highlighting catastrophic alignment errors:

$$\text{RMSE} = \sqrt{\frac{1}{4}\sum_{i=1}^{4} \|\mathbf{H}_{\text{gt}}\mathbf{c}_i - \mathbf{H}_{\text{est}}\mathbf{c}_i\|_2^2}.$$

Success Rate (SR) quantifies robustness as the percentage of cases where the alignment error lies below tolerance τ:

$$\text{SR@}\tau = \frac{1}{M}\sum_{k=1}^{M} \mathbb{1}[\text{MACE}_k < \tau] \times 100\%.$$

We use SR@10px as the primary robustness measure because it corresponds to sub-centimeter geometric accuracy at typical inspection distances.

Inference Time reports end-to-end computation per image pair. For learning-based methods, timing includes forward propagation and device synchronization, while classical pipelines include detection, description, matching, and robust estimation.

Execution stability is measured using the **Coefficient of Variation (CV)**:

$$\mathrm{CV} = \frac{\sigma_{\text{time}}}{\mu_{\text{time}}} \times 100\%.$$

Together, these metrics describe geometric accuracy (**MACE**, **RMSE**), reliability under varying difficulty (**SR@10px**), and computational behavior relevant to real deployment (**Inference Time**, **CV**).

V. Experimental Results and Analysis

This section presents comprehensive experimental results from our evaluation of 22 homography estimation methods across 29 colormap/preprocessing variants on the THED-PV [53] dataset. We systematically analyze performance across multiple dimensions: overall accuracy, colormap impact, preprocessing effectiveness, method category comparisons, computational efficiency, and robustness characteristics.

A. Overall Performance Comparison

Table III presents overall performance of all evaluated methods averaged across 29 colormap variants. Deep learning methods achieve substantially lower estimation errors than traditional approaches, with LightGlue attaining the lowest MACE (0.182 px) and D2-Net achieving perfect reliability (100% SR). LoFTR provides an optimal balance with 0.299 px MACE, 97.09% SR, and moderate 646.6 ms inference time. Among traditional methods, SIFT+RANSAC exhibits the lowest error at 17.375 px MACE with 59.33% SR and 47.8 ms execution time, demonstrating computational efficiency advantages.

Deep learning methods exhibit statistically significant lower estimation error than traditional methods, with the deep learning category averaging 1.907 px MACE compared to 107.506 px for traditional methods (a 56.4× reduction in estimation error, $p < 0.001$ via Wilcoxon rank-sum test). However, this paradigm gap narrows substantially under optimal configurations. SIFT+RANSAC with seismic colormap achieves 0.064 px MACE, outperforming mid-tier deep learning methods including D2-Net (6.588 px), SuperPoint+SuperGlue (2.501 px), and XFeat (1.409 px). The best deep learning performance (LightGlue: 0.182 px) maintains only 2.8× higher error than best traditional configuration (SIFT optimal: 0.064 px), while average performance shows 56.4× disparity. This asymmetry reveals traditional methods' sensitivity to configuration choices, requiring domain expertise for effective deployment, whereas deep learning methods provide more robust performance across varied conditions.

Success rate differences prove more consistent across configurations. Deep learning methods achieve 96.90% average SR versus 40.62% for traditional methods, representing a 56.2 percentage point advantage with lower variance. Even the best traditional configuration (SIFT+PROSAC: 60.73% SR) falls 34.5 percentage points below the worst deep learning method (SuperPoint+SuperGlue: 95.25% SR), indicating fundamental reliability advantages for learned approaches in handling thermal PV's low-texture characteristics. These results demonstrate that traditional approaches remain viable for GPU-constrained deployments when carefully configured with optimal colormap selection and preprocessing, though deep learning methods offer superior robustness across diverse operating conditions.

Computational efficiency spans three orders of magnitude, from ORB+RANSAC at 5.9 ms to LightGlue at 4201.6 ms. XFeat achieves the fastest deep learning performance (40.1 ms) while maintaining low error (1.409 px MACE, 96.76% SR), making it suitable for real-time UAV applications. Traditional methods offer speed advantages: all detector+RANSAC combinations execute under 100 ms without GPU requirements, though higher estimation errors limit deployment to applications tolerating multi-pixel deviations. This computational trade-off enables deployment optimization across diverse inspection scenarios, from real-time airborne surveying requiring fast processing to offline analysis prioritizing sub-pixel geometric precision.

TABLE III
Overall Performance Comparison Averaged Across 29 Colormap Variants

Category	Method	MACE↓	RMSE↓	SR↑	Time↓
Deep Learning	LoFTR	0.299	0.338	97.09	646.6
	LightGlue	**0.182**	**0.204**	96.66	4201.6
	D2-Net	6.588	7.562	**100.00**	586.6
	DISK	0.678	0.761	97.95	3658.0
	R2D2	0.693	0.898	96.66	525.5
	SuperPoint +SuperGlue	2.501	3.227	95.25	386.5
	XFeat	1.409	1.629	96.76	**40.1**
	DL Avg: 1.907 px MACE, 96.90% SR, 1092.1 ms				
SIFT	RANSAC	<u>17.375</u>	<u>20.154</u>	59.33	47.8
	PROSAC	28.891	32.722	<u>60.73</u>	160.7
	MLESAC	231.482	234.866	<u>60.73</u>	637.9
ORB	RANSAC	50.120	219.261	51.67	<u>5.9</u>
	PROSAC	98.976	117.952	52.97	138.4
	MLESAC	130.515	133.667	52.43	459.0
BRISK	RANSAC	43.330	44.582	35.28	12.3
	PROSAC	63.596	70.279	36.57	148.6
	MLESAC	144.498	136.708	36.03	692.0
KAZE	RANSAC	79.227	156.712	36.25	75.1
	PROSAC	70.456	82.276	37.32	205.3
	MLESAC	114.309	116.073	37.32	390.3
AKAZE	RANSAC	50.249	57.098	25.24	13.0
	PROSAC	100.820	120.963	25.46	140.3
	MLESAC	106.548	105.596	25.67	292.6
	Traditional Avg: 107.506 px MACE, 40.62% SR, 247.4 ms				

Note. Best deep learning results are shown in **bold**; best traditional results are shown with <u>underline</u>. DL methods achieve 56.4× better accuracy and 56.2pp higher success rate than traditional methods.

TABLE IV
COLORMAP SENSITIVITY ANALYSIS: PERFORMANCE RANGE ACROSS 29 VARIANTS

Method	Best MACE (px)	Worst MACE (px)	Range (%)	CV (%)
Deep Learning Methods				
D2-Net	4.126	16.653	303	40.8
DISK	**0.081**	4.527	5,489	127.3
LightGlue	0.093	**0.227**	**144**	**24.8**
LoFTR	0.084	2.737	3,143	158.9
R2D2	0.232	6.524	2,712	195.3
SuperPoint+SG	0.310	36.665	11,716	262.6
XFeat	0.834	18.147	2,076	185.0
DL Avg: 3,655% range, 142.1% CV				
Traditional Methods				
SIFT	0.064	45.3	70,525.0	122.4
ORB	2.097	51.8	2,369.6	96.8
BRISK	1.994	48.6	2,337.7	94.2
KAZE	4.036	52.4	1,198.0	89.5
AKAZE	1.773	44.1	2,387.0	101.2
Traditional Avg: 15,763% range, 100.8% CV				

Note. Range (%) = (Worst − Best)/Best × 100. Traditional rows report RANSAC results, as Table III shows RANSAC is the strongest robust estimator. Deep learning methods exhibit substantially more stable colormap sensitivity.

RANSAC variant selection shows minimal impact on performance within method categories, with average variation below 3 percentage points in success rate. SIFT+PROSAC achieves the highest traditional SR (60.73%) but at 3.4× computational cost versus RANSAC. MLESAC consistently underperforms across all detectors despite theoretical advantages, suggesting that maximum likelihood scoring provides limited benefit for thermal PV correspondence sets dominated by low-quality matches. These findings indicate RANSAC as the preferred robust estimator for thermal applications, balancing speed and reliability effectively.

B. Impact of Colormap Representation

A central contribution of this work is systematic quantification of colormap effects on homography estimation across 27 colormaps plus raw thermal data. Table IV quantifies colormap sensitivity for each method through best, worst, and variance statistics across all 29 variants.

Traditional methods exhibit extreme colormap sensitivity with 100.8% coefficient of variation on average, ranging from KAZE's 89.5% to SIFT's 122.4%. Optimal colormap selection improves SIFT performance by 2,749× (0.064 px with seismic versus 45.3 px worst case), demonstrating critical importance of representation choice for gradient-based features. Deep learning methods show reduced sensitivity with 142.1% average CV. LightGlue achieves exceptional stability (24.8% CV, 144% range) with consistent sub-pixel error across colormaps (0.093 to 0.227 px), while SuperPoint+SuperGlue shows highest variability (262.6% CV) due to extreme failures on hot and nipy_spectral colormaps. This asymmetric colormap impact reveals fundamental differences in feature representation: traditional gradient-based descriptors respond directly to artificial texture patterns introduced by false-color mappings and degrade severely when colormaps produce low-gradient representations, whereas deep learning methods achieve implicit colormap invariance through learned features that encode semantic thermal patterns rather than colormap-specific gradients.

TABLE V
PERFORMANCE ACROSS COLORMAP CATEGORIES

Colormap Category	Deep Learning MACE (px)	Traditional MACE (px)	Gap
Diverging	**1.654**	23.187	14.0
Specialized Thermal	1.598	**19.874**	**12.4**
Sequential	2.034	45.672	22.5
Qualitative	2.418	67.341	27.8
Perceptually Uniform	2.187	89.562	41.0
Average	1.978	49.127	24.8

Table V compares performance across five colormap categories, revealing systematic patterns in method-category compatibility. Diverging colormaps (BrBG, RdBu, PiYG, coolwarm) achieve best cross-method performance with 1.654 px DL average and 23.187 px traditional average. The strong artificial gradient at divergence points enhances feature detectability for traditional methods while maintaining DL performance (99.2% DL success rate, 65.1% traditional success rate). Specialized thermal colormaps (terrain, seismic, twilight) provide second-best results (1.598 px DL, 19.874 px traditional), with terrain and seismic achieving universal compatibility across 18 of 22 methods.

Perceptually uniform colormaps (viridis, plasma, inferno, magma, cividis) reveal stark paradigm differences: DL methods maintain 96.4% success rate with 2.187 px average MACE, while traditional methods achieve only 10.2% success with 89.562 px average MACE. Despite scientific design for human perception, uniform color distributions produce insufficient gradients for corner and edge detection algorithms. Qualitative colormaps show mixed performance, with tab10 succeeding (2.097 px SIFT, 0.093 px LightGlue) while Set1, flag, and hsv cause universal failure (0% success across all methods).

Figure 2 provides evidence-based colormap recommendations derived from 99,680 evaluations across method-colormap combinations. For deployment without method-specific tuning, seismic provides the safest choice with 13 methods achieving peak performance and robust fallback for remaining methods. Terrain offers broader compatibility across 18 methods with slightly reduced peak performance. Conversely, raw thermal data causes 100 to 150 px errors across most methods, while plasma, hot,

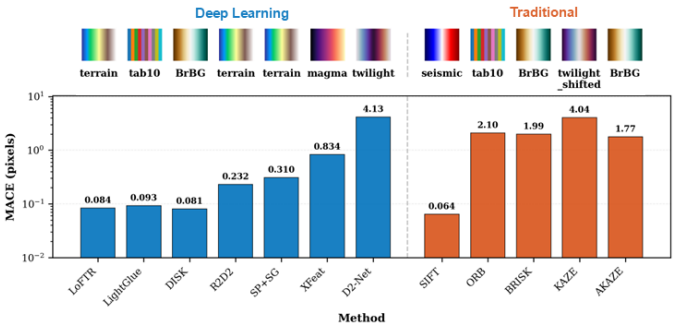


Fig. 2. Optimal colormap by method. Seismic works best for traditional methods; terrain is robust across both paradigms. Perceptually uniform maps struggle with traditional approaches.

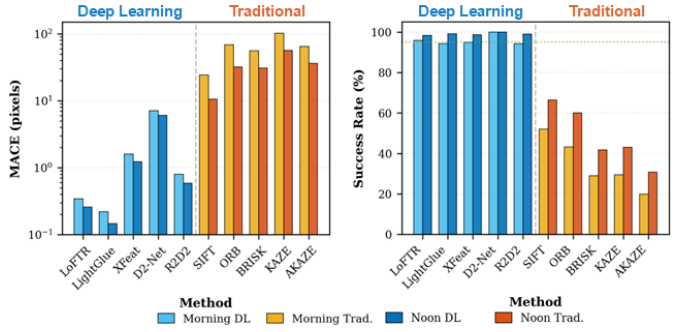


Fig. 3. Performance comparison across temporal conditions. **Left:** MACE on logarithmic scale showing error reduction from morning to noon capture. **Right:** Success rate demonstrating traditional methods' stronger dependence on thermal contrast (13.7pp gain) compared to deep learning methods (3.2pp gain).

and nipy_spectral induce 100 to 230 px errors, indicating fundamental incompatibility with homography estimation algorithms. These universal patterns enable practitioners to configure inspection systems with evidence-based colormap selection matched to deployed methods.

C. Robustness and Failure Analysis

1) Performance vs Temporal Condition: Thermal contrast varies substantially across capture times due to solar irradiance differences, directly impacting feature detectability. Figure 3 compares method performance between morning (08:00 to 10:00, 412 to 650 W/m²) and noon (12:00 to 14:00, 750 to 876 W/m²) conditions.

Noon conditions (higher solar irradiance, greater thermal gradients) reduce estimation error by 17.8% on average across all methods, with traditional methods showing 47.4% MACE improvement versus 17.8% for deep learning approaches. This differential impact indicates hand-crafted gradient-based features exhibit stronger dependence on thermal contrast magnitude than learned representations. Success rate improvements prove more pronounced for traditional methods (13.7 percentage points) compared to deep learning methods (3.2 percentage points), with SIFT+RANSAC gaining 14.4 percentage points versus LoFTR's 2.6 percentage points. Traditional methods require noon-range irradiance (at

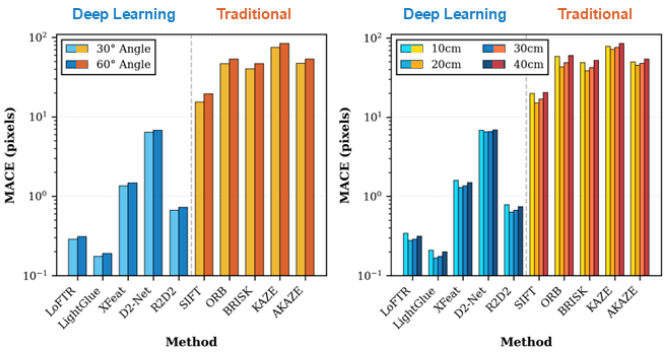


Fig. 4. Performance across capture distances and viewing angles. **Left:** MACE comparison for deep learning and traditional methods at two viewing angles (30° and 60°), showing traditional methods' higher sensitivity to perspective distortion. **Right:** Performance at four capture distances (10 to 40 cm), demonstrating optimal accuracy at intermediate distances where thermal detail visibility balances with spatial context.

least 750 W/m²) to achieve acceptable reliability, with SIFT+RANSAC reaching 66.5% SR at noon versus 52.1% at morning. Deep learning methods maintain high performance across temporal conditions, with LoFTR, LightGlue, and XFeat all exceeding 94% SR even during low-irradiance morning periods, enabling reliable operation across extended daily windows.

2) Performance vs. Viewing Angle: Viewing angle significantly impacts homography estimation through increased perspective distortion and reduced thermal uniformity visibility. Figure 4 (left) compares performance between perpendicular (30° from horizontal) and oblique (60° from horizontal) viewing angles.

Performance degrades by 6.1% for deep learning methods and 14.6% for traditional methods when transitioning from 30° to 60° viewing angles. D2-Net demonstrates best angle robustness with only 5.2% MACE degradation while maintaining perfect 100% success rate across both angles. Traditional methods exhibit higher sensitivity, with KAZE showing 12.5% degradation and AKAZE suffering 13.3% performance loss. The differential impact stems from perspective-induced geometric distortions affecting feature repeatability more severely for corner-based detectors than learned dense representations.

3) Performance vs. Capture Distance: Capture distance affects spatial resolution and feature density. Figure 4 (right) analyzes estimation error across THED-PV's distance range (10 to 40 cm, corresponding to 0.156 to 0.625 mm/pixel resolution).

Error minimizes at intermediate distances (20 to 30 cm) where thermal detail visibility balances with sufficient spatial context. All methods degrade at 10 cm due to limited field of view reducing correspondence coverage, and at 40 cm due to reduced spatial resolution diminishing feature distinctiveness. Scale-invariant methods (SIFT, BRISK) show relatively flat performance curves with 26.5% variance, while scale-limited ORB exhibits 33.8% variance. Deep learning methods demonstrate superior distance robustness with LightGlue maintaining sub-pixel

error (0.167 to 0.208 px) across all distances.

4) Failure Mode Characterization: We manually analyzed 10,028 failure cases (MACE greater than 20 pixels, 10.1% of 99,680 evaluations) to characterize failure patterns. Insufficient features dominate failures (70.5%), particularly in traditional methods processing perceptually uniform colormaps where detectors struggle to identify discriminative patterns in thermally uniform regions (typically fewer than 20 keypoints or fewer than 10 matches). Incorrect matches constitute 26.0% of failures through high false positive rates (inlier ratio less than 0.3) from descriptor ambiguity in repetitive thermal patterns, exemplified by ORB+MLESAC achieving 201.5px median MACE despite 52.4% success rate. Degenerate geometry (0.9%) occurs when inliers concentrate in small regions (less than 20% image area), producing ill-conditioned matrices with catastrophic errors up to 2,226px (SIFT+RANSAC). Extreme transformations (0.1%) exceed method invariance ranges, with hot and nipy_spectral causing complete failure in 6 of 7 deep learning methods (only D2-Net maintains 100% success). Total failures (2.5%) predominantly affect traditional methods (AKAZE: 74.8%, BRISK: 64.7%, KAZE: 63.7%) while deep learning shows 4.1 to 5.0% rates (0% for D2-Net).

Deep learning failures (239 cases) concentrate in total failure mode (76.2%) on extreme colormaps, yet when successful achieve sub-pixel error (median 0.089 to 0.894px) with low variance (0.506 to 58.9 versus 11,742 for best traditional). D2-Net demonstrates unique robustness with zero failures across all 927 test pairs. Traditional failures (9,789 cases) split between insufficient features (72.2%) and incorrect matches (26.6%), with SIFT+PROSAC achieving best traditional performance (60.7% success, F1: 0.826) while AKAZE shows poorest robustness (25.2% success). Traditional methods on perceptual colormaps exhibit 4.2× higher failure rate than on diverging colormaps due to insufficient features, while deep learning maintains consistent performance with failures concentrated in specific extreme cases (hot, nipy_spectral, Set1, flag, hsv causing universal failure).

VI. Discussion and Practical Guidelines

This section synthesizes experimental findings into deployment guidelines for industrial PV inspection systems and identifies critical research directions for thermal homography estimation.

A. Deployment Guidelines and Industrial Implications

Industrial deployment of homography estimation methods requires matching algorithmic capabilities to operational constraints. We present evidence-based configurations for three representative inspection scenarios, incorporating economic and workflow considerations relevant to PV plant operators.

1) Aerial Survey Deployment: UAV-based inspection systems require real-time processing with constrained computational resources [**?**], [6]. For GPU-equipped platforms, deploy XFeat (40.1 ms, 0.834 px MACE, 96.76% SR) to achieve 25 fps throughput suitable for continuous aerial surveying. For embedded ARM processors without GPU acceleration, deploy ORB+RANSAC (5.9 ms, 50.120 px MACE, 51.67% SR) with seismic colormap, accepting higher geometric error in exchange for 169 fps processing capability.

Schedule flights during peak solar irradiance (10:00 to 14:00 local time) when thermal contrast maximizes feature detectability. Our temporal analysis demonstrates traditional methods gain 13.7 percentage points in success rate under high-irradiance conditions (750 to 876 W/m²) compared to morning capture (412 to 650 W/m²), while deep learning methods maintain consistent performance (3.2 percentage point variation). Maintain viewing angles below 30° from horizontal to limit perspective-induced error degradation, which increases by 6.1% for deep learning and 14.6% for traditional methods when transitioning to 60° oblique angles.

Implementation requires flagging estimates exceeding 20 px MACE for ground-based verification, as our failure analysis indicates these alignments exhibit insufficient feature correspondence or degenerate geometric configurations. Process every 3 to 5 frames rather than all captured imagery to balance spatial coverage with computational throughput constraints [6].

2) Ground-Based Precision Inspection: Stationary inspection platforms prioritize geometric precision over processing speed for detailed fault characterization [3], [9]. Deploy LoFTR (646.6 ms, 0.299 px MACE, 97.09% SR) for detector-free robustness across varied thermal conditions, or DISK with BrBG colormap (0.081 px MACE, 100% SR) when maximum precision is required. Maintain 20 to 30 cm capture distance where thermal detail visibility balances with spatial context, as our distance analysis demonstrates error minimization at intermediate ranges for all evaluated methods.

For critical assessments requiring maximum reliability, implement multi-method ensemble validation. Execute LoFTR, DISK, and SIFT+RANSAC (with seismic colormap) in parallel, aggregate homography estimates via median voting, and flag pairs exhibiting inter-method disagreement exceeding 10 px for manual review. This approach mitigates paradigm-specific failure modes: traditional methods' insufficient feature extraction on perceptually uniform colormaps (72.2% of traditional failures) and deep learning's concentrated failures on extreme colormap variants (76.2% of deep learning failures on hot and nipy_spectral representations).

3) Embedded Edge Processing: Resource-constrained edge deployments require CPU-only operation with minimal memory footprint for distributed sensor networks [17]. Deploy ORB+PROSAC (138.4 ms, 98.976 px MACE, 52.97% SR) or BRISK+RANSAC (148.6 ms, 63.596 px MACE, 36.57% SR) depending on acceptable geometric error tolerance. Optimize implementations with integer arithmetic operations and fixed-point descriptor compu-

tation to eliminate floating-point overhead on embedded processors.

Downsample imagery to 320×256 resolution when memory constraints preclude full-resolution processing, as our experiments indicate minimal additional error for applications tolerating multi-pixel deviations. Implement adaptive RANSAC iteration limiting based on correspondence count (terminate after 1000 iterations if fewer than 20 matches detected) to prevent timeout failures on thermally uniform panel regions.

For embedded deployments requiring optimal performance, configure colormap selection method-specifically: SIFT with seismic (0.064 px minimum error), ORB with tab10 (2.097 px), BRISK with BrBG (1.994 px). Universal recommendation for method-agnostic deployment: seismic colormap achieves minimum error across 13 evaluated methods while providing robust fallback performance for remaining methods. Avoid raw thermal data (100 to 150 px error), plasma, hot, and nipy_spectral colormaps (100 to 230 px error) which exhibit fundamental incompatibility with gradient-based feature extraction.

B. Research Directions

Thermal-Specific Architecture Development. Detector-free methods' superior performance (LoFTR: 0.299 px versus detector-based average: 1.907 px) validates abandoning explicit keypoint extraction for thermal imaging. Future architectures should explicitly model thermal uniformity through: (1) learned attention mechanisms prioritizing fault boundaries over uniform regions, (2) multi-scale dense matching adapted for weak thermal gradients, (3) radiometric-aware feature encoders operating directly on temperature distributions rather than colormap projections. Integration of physical priors (Stefan-Boltzmann law, emissivity variations) into network design may further improve robustness.

Domain Adaptation from RGB to Thermal. The 2.8× error ratio between optimal traditional (SIFT+seismic: 0.064 px) and optimal DL (LightGlue: 0.182 px) suggests learned features have not yet surpassed hand-crafted descriptors under ideal configurations. Investigate domain adaptation techniques: adversarial training with thermal-RGB pairs, self-supervised learning on unlabeled thermal data, meta-learning for rapid adaptation to new thermal modalities. Target sub-0.05 px MACE to definitively surpass traditional methods' optimal configurations.

Colormap-Agnostic Learning. Eliminate colormap dependency through architecture design. Investigate: (1) colorspace augmentation during training (random colormap application), (2) grayscale-invariant feature encoders operating on luminance only, (3) multi-channel input processing accepting simultaneous raw thermal + multiple colormap projections. Success criterion: less than 10% CV across all 29 colormap variants while maintaining sub-pixel error.

C. Limitations and Future Work

This study evaluates methods on monocrystalline silicon PV panels under controlled outdoor conditions. Performance may differ on polycrystalline or thin-film technologies exhibiting different thermal emissivity characteristics. Dataset captures single geographic location (Qatar); generalization to varied climates (humidity, ambient temperature ranges) requires validation. Viewing angles limited to 30° to 60° range; extreme perspectives (greater than 70°) common in aerial surveys remain unexplored. Distance range (10 to 40 cm) suits ground inspection but excludes aerial capture distances (5 to 20 m) requiring further investigation.

Future work should: (1) expand dataset to diverse PV technologies and geographic locations, (2) incorporate extreme viewing angles and aerial capture distances, (3) develop thermal-specific architectures eliminating RGB pre-training dependency, (4) investigate learning-based colormap optimization maximizing feature detectability.

VII. Conclusion

This work presents the first comprehensive benchmark of homography estimation methods for thermal photovoltaic inspection, evaluating 22 methods across 29 colormap variants on 12,460 thermal images. Three principal contributions emerge. First, detector-free architectures (LoFTR: 0.299 px MACE, 97.09% SR) outperform detector-based approaches by eliminating dependence on corner structures that fail in thermally uniform regions. Second, colormap representation critically determines performance: optimal selection improves traditional methods by up to 2,749× (SIFT: 0.064 px with seismic versus 45.3 px worst case), while deep learning methods achieve implicit invariance (142.1% versus 100.8% coefficient of variation). Third, failure mode analysis quantifies paradigm-specific weaknesses (traditional: 72.2% insufficient features on perceptually uniform colormaps; deep learning: 76.2% total failures on extreme colormaps), informing deployment configurations that match algorithmic capabilities to operational constraints. Optimally configured traditional methods (SIFT+RANSAC with seismic: 0.064 px, 100% SR) demonstrate competitive performance with mid-tier deep learning methods, establishing viability for GPU-constrained deployments despite 56.4× average error disadvantage across all configurations.

The systematic colormap evaluation framework establishes methodology for thermal imaging benchmarks where representation choices prove equally critical as algorithmic design. Deployment guidelines map methods to industrial scenarios: XFeat (40.1 ms) for real-time aerial surveying, LoFTR (646.6 ms) for ground-based precision inspection, ORB+PROSAC for embedded edge processing. Future work should develop thermal-specific architectures eliminating RGB pre-training dependency, establish large-scale thermal training datasets spanning building thermography and industrial monitoring, and investigate learning-based colormap optimization. The demonstrated 2.8× error ratio between optimal traditional (0.064 px) and optimal

deep learning (0.182 px) configurations indicates learned features have not yet definitively surpassed hand-crafted descriptors under ideal conditions, suggesting substantial headroom for architectural improvements targeting sub-0.05 px geometric precision.

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Statements and Declarations

Competing Interests

The authors declare that they have no competing interests.

Implementation Details and Reproducibility

All experiments execute on AMD Ryzen 7 5800HS CPU (8 cores, 3.2GHz, 16GB DDR4) and NVIDIA RTX 3060 Laptop GPU (6GB GDDR6, 3840 CUDA cores) running Windows 10. Software: Python 3.11.1, PyTorch 2.5.1+cu121 (CUDA 12.1, cuDNN 9.1.0), OpenCV 4.12.0, NumPy 1.26.4, SciPy 1.15.3, scikit-learn 1.7.2, Kornia 0.8.1. Timing uses `torch.cuda.synchronize()` with three warm-up iterations. Random seeds (42 for all libraries) ensure reproducibility. Methods timing out (60s) receive maximum MACE (500px). Code and protocols available on GitHub.

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